

NAWSHIN AHANA (AHANA)

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EDUCATION

University of Pennsylvania | Philadelphia, PA

B.A., Mathematics and Computational Biology

B.A., Cultures and Societies of the Middle East and North Africa (CAS MENA) -- Honors Program

Relevant Coursework: Pricing Policy, Personal Finance, Analysis, Causal Inference, Artificial Intelligence (Back-End Functions), Modern Data Mining.

Honors & Awards: Mayor's Scholar; Steinhardt Scholar; Pioneer Academics Scholar; Quest for Excellence STEM Award

ANALYTICS & RESEARCH EXPERIENCE

Wharton Sports Analytics & Business Initiative (WSABI), University of Pennsylvania — Philadelphia, PA

Quantitative Research Fellow | Mar — Aug 2026

- Analyze large-scale Formula 1 race datasets to evaluate strategic decision-making, competitive dynamics, and trade-offs under uncertainty using Python and R
- Build Bayesian statistical models and Markov Decision Processes to quantify uncertainty and assess race strategy performance and minimize risky response
- Currently working on analytics research on Perelli tire and their functionality

Penn Aerospace Club, University of Pennsylvania — Philadelphia, PA

Mechanical Payload Team | *Penn High-Altitude Balloon Team* | Feb — Jul 2026

- Engineered and assembled payload and structural components as part of the mechanical build team for the Penn High-Altitude Balloon Team
- Conducted pre-flight testing and payload integration to ensure structural reliability and flight readiness in near-space conditions, collaborating with a multidisciplinary engineering team on mission design and launch preparation

U.S. National Science Foundation (NSF), Mountain Lake Biological Research Station — Mountain Lake, VA

Field Researcher | May 2025

- Conducted field study on anti-herbivore defense traits in *Medicago lupulina*, designing research methodology and executing data collection
- Applied statistical and computational methods to identify evidence-based patterns in plant defense responses; developed parallel framework translating ecological findings into potential applications

Sgourakis Lab, Children's Hospital of Philadelphia (CHOP) — Philadelphia, PA

Undergraduate Research Assistant | Jan — Nov 2024

- Supported translational cancer research by preparing and validating lentiviral constructs for tumor-cell transduction and CAR T-Cell engineering
- Executed molecular biology workflows including plasmid preparation, sequencing validation, mammalian cell transduction, and qPCR; maintained experimental reproducibility
- Synthesized experimental results into technical presentations, communicating findings to senior scientists.

Pioneer Academics

Research Fellow | Jun — Aug 2022

- Conducted independent neuroscience research through Pioneer Academics under the supervision of Dr. Bradley Gibson (University of Notre Dame), focusing on interoceptive awareness and its cognitive-neuroscientific mechanisms. Gained experience in scientific writing, experimental design concepts, and interpreting psychological/neuroscience research.

CONSULTING & CORPORATE EXPERIENCE

Capital One × UPenn Consult for America Case Competition

Participant | Apr 2026

- Partnered with a 4-person team to develop an international growth strategy for Capital One, leveraging market sizing, competitive analysis, and structured problem-solving to evaluate expansion opportunities.
- Built a 20-slide executive-ready recommendation deck and presented to Corporate Strategy leaders, translating insights into a clear, actionable roadmap for global growth.

Boston Consulting Group (BCG) — Chicago, IL

Bridge to Consulting Program | May 2025

- Competitively selected for BCG's early-career consulting program emphasizing structured problem-solving, strategic analysis, and client-focused communication.
- Applied core consulting frameworks to real-world business cases, developing data-backed recommendations through collaborative team-based work.
- Worked with BCG consultants and peers to evaluate a market entry and pricing strategy case, conducting market sizing, competitive assessment, and pricing analysis to shape actionable recommendations.

Sheppard Financial Group — Philadelphia, PA

Financial Advisor Intern | May – Jun 2025

- Conducted client needs analysis and developed personalized financial strategies aligned with long-term goals, translating complex financial concepts into clear client-facing recommendations
- Built foundational expertise in financial services, client relationship management, and solution-oriented problem solving under licensed advisors

CVS Health — Philadelphia, PA

Spring STEM Student Mentorship Program | Mar – Apr 2025

- Selected for competitive mentorship program with Digital, Analytics, and IT teams exploring data-driven solutions to healthcare challenges
- Collaborated on cross-functional projects applying analytics and technology strategy to improve operational efficiency and patient outcomes

TEACHING & LEADERSHIP

Wharton Moneyball Academy, University of Pennsylvania | Philadelphia, PA

Teaching Assistant | Jun – Jul 2026

- Mentored 10 students through independent sports analytics projects (tennis performance, NBA injury recovery), providing guidance on statistical modeling, coding, and data interpretation
- Supported 75 students daily by troubleshooting R code, explaining statistical concepts, and guiding applied data analysis assignments
- Communicated technical statistical concepts to students with varied quantitative backgrounds.

Wharton School, University of Pennsylvania | Philadelphia, PA

Teaching Assistant, Essentials of Personal Finance | Aug 2025 – Present

- Co-deliver financial education curriculum to 50+ high school students from underserved communities, translating concepts including interest, credit, taxation, and financial decision-making into accessible instruction
- Facilitate discussions and provide individualized feedback, adapting explanations to different levels of quantitative and financial literacy
- Collaborate with MBA students and program leadership to refine curriculum delivery and expand reach of financial education initiatives

VOLUNTEER & COMMUNITY LEADERSHIP

Philomathean Society, University of Pennsylvania | Philadelphia, PA

Tea Chair | Aug 2026 – Present

- Coordinate with UPenn professors and faculty to organize enriching talks for Penn members
- Lead talks and facilitate student engagement and learning

Matriculate, University of Pennsylvania | Philadelphia, PA

Advising Fellow | 2024 – Present

- Provide comprehensive one-on-one guidance to 3 first-generation, low-income high school students through the full college application lifecycle
- Lead essay brainstorming and drafting, ensuring authentic, compelling personal narratives
- Develop tailored financial aid strategies including FAFSA/CSS Profile navigation and scholarship identification, enabling informed decisions between offers of admission

College of Arts & Sciences, University of Pennsylvania | Philadelphia, PA

Peer & MAP Advisor | May 2025 – 2026 Sep

- Mentor 36 first-year students on academic planning, course selection, campus engagement, and university resources
- Facilitate goal-setting and coordinate communication between students and pre-major advisors

Zeta Tau Alpha, University of Pennsylvania | Philadelphia, PA

Activities Director | Dec 2025 – Feb 2026 | *Assistant Activities Director* | Feb – Dec 2025

- Spearheaded end-to-end execution of large-scale events for 200+ attendees, managing vendor contracts, venue logistics, and on-site production to ensure seamless experiences
- Directed event budgets, meticulously tracking expenditures and reallocating funds to maintain strict financial compliance while maximizing member engagement
- Balanced competing priorities across multiple stakeholders (executive board, venue staff, chapter members), delivering high-quality programming within budget and operational constraints

Pennsylvania Hospital | Philadelphia, PA

Volunteer | Oct 2024 – Sep 2026

- Provide direct pre- and post-procedure support to patients, families, and nursing teams
- Demonstrate composure and adaptability in fast-paced environment, rapidly reprioritizing tasks to align with shifting physician schedules and acute patient needs

SKILLS & ADDITIONAL INFORMATION

Technical: Python, R, Data Analysis & Visualization, Microsoft Office Suite, Web Development

Quantitative: Bayesian Statistics, Statistical Modeling, Markov Decision Processes, Data Visualization, Research Design

Healthcare: Clinical Research, CAR T-Cell Engineering, Virus Production, Patient Care, Healthcare Operations

Affiliations: Philomathean Society UPenn; UPenn Women's Rugby; Minds Matter; Zeta Tau Alpha; QuestBridge; Matriculate

Languages: English (Fluent), Bengali (Native), Hindi (Proficient)